Exploratory Data Analysis: The Beauty Industry & The "Luxury Marker"

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Data Analysis with Python and SQL Winter Quarter 2026

CHICAGO BOOTH
The University of Chicago Booth School of Business

The Beauty Industry Paradox

- DO HIGHER PRICES GUARANTEE SUPERIOR QUALITY?
- IS PRICE A SIGNAL OF PERFORMANCE OR A BYPRODUCT OF MARKETING?

- **AUDIENCE:** CONSUMERS NAVIGATING MARKETING "FLUFF" AND BRAND MANAGERS SEEKING MARKET POSITIONING
- ***MARKETING FOCUS:*** *PRODUCT COPY LENGTH AND GREEN CLAIMS*
- **MAKEUP API:** +900 PRODUCTS FROM +50 BRANDS (AFTER CLEANING)

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=== Query 2: Missing Data by Column ===
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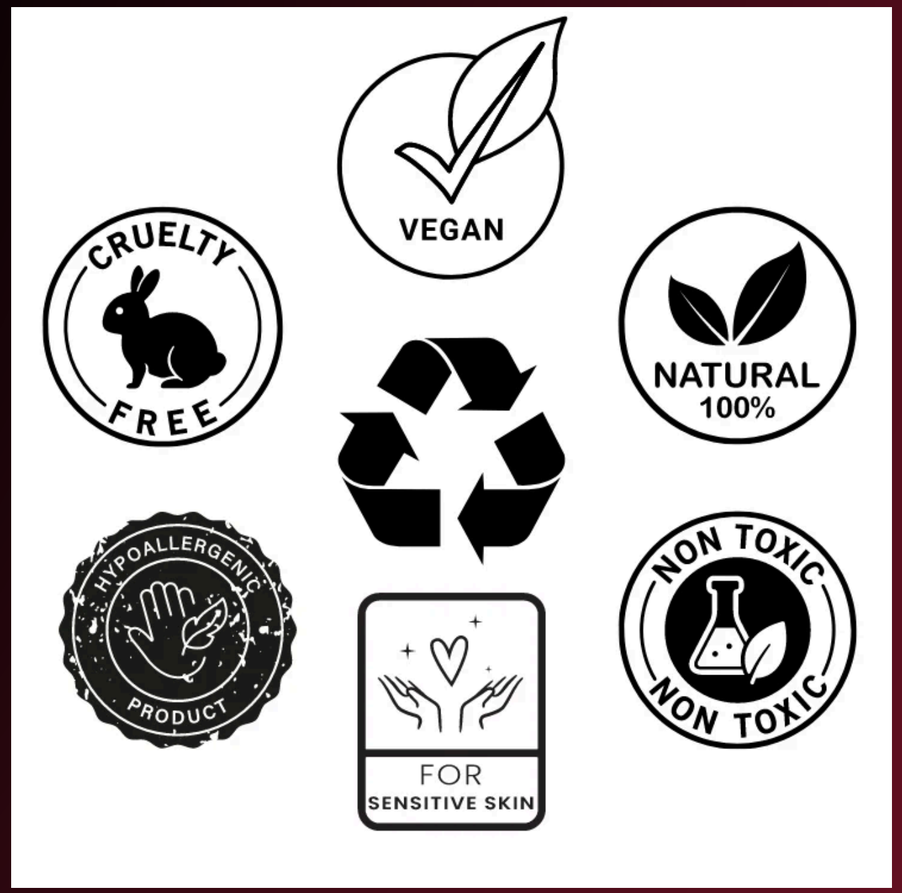
	column_name	missing_count	missing_pct
0	rating	591	63.5
1	price	54	5.8
2	description	25	2.7
3	brand	12	1.3
4	product_type	0	0.0

Our Hypotheses

- PRODUCTS MARKETED WITH "GREEN" OR SUSTAINABILITY RELATED TERMS WILL RECEIVE HIGHER RATINGS AND POTENTIALLY COMMAND HIGHER PRICES DUE TO INCREASING CONSUMER DEMAND FOR SUSTAINABLE COSMETICS
    - PRICE–QUALITY GAP
    - DESCRIPTION PREMIUM
    - ARE GREEN CLAIMS A STANDARD OR LUXURY?



Finding I: The Price-Quality Myth

price_tier	avg_rating
1_Budget (<$10)	4.31
2_Mid-range ($10–19)	4.29
3_Premium ($20–34)	4.43
4_Luxury ($35+)	4.33

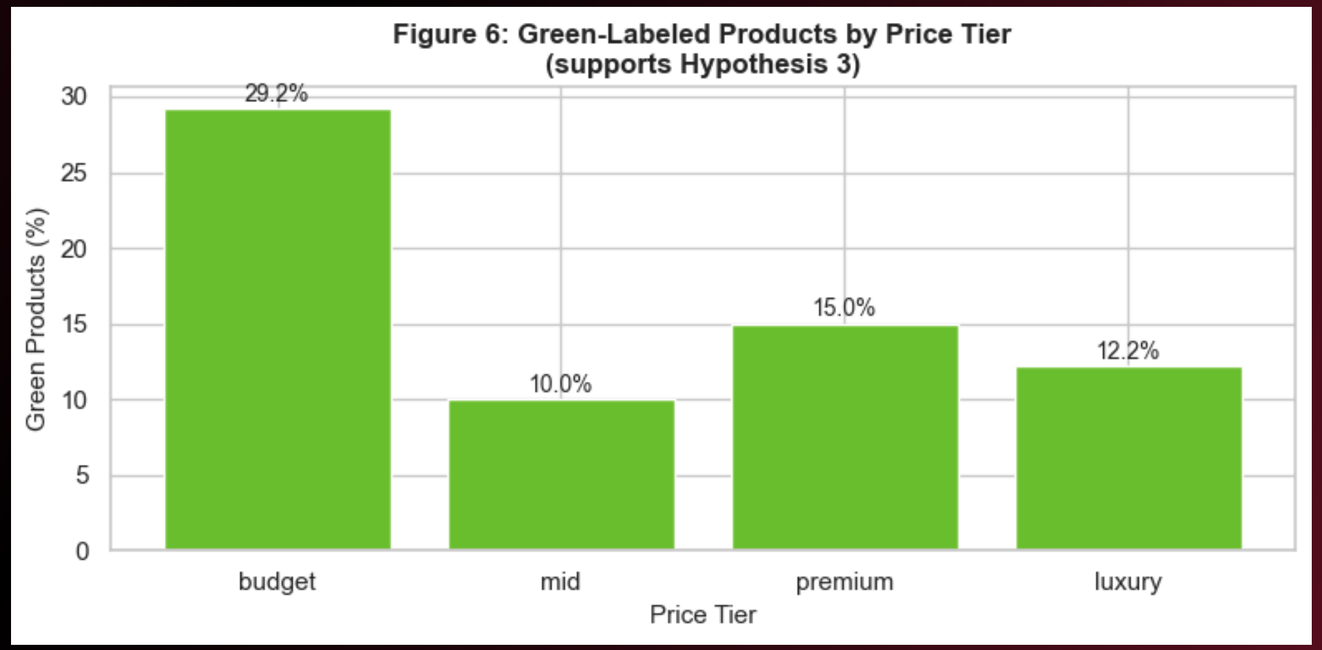
- Our expectation: Luxury or “better” products would have significantly higher ratings.
- Reality: Consistent ratings across different tiers. Weakly correlated

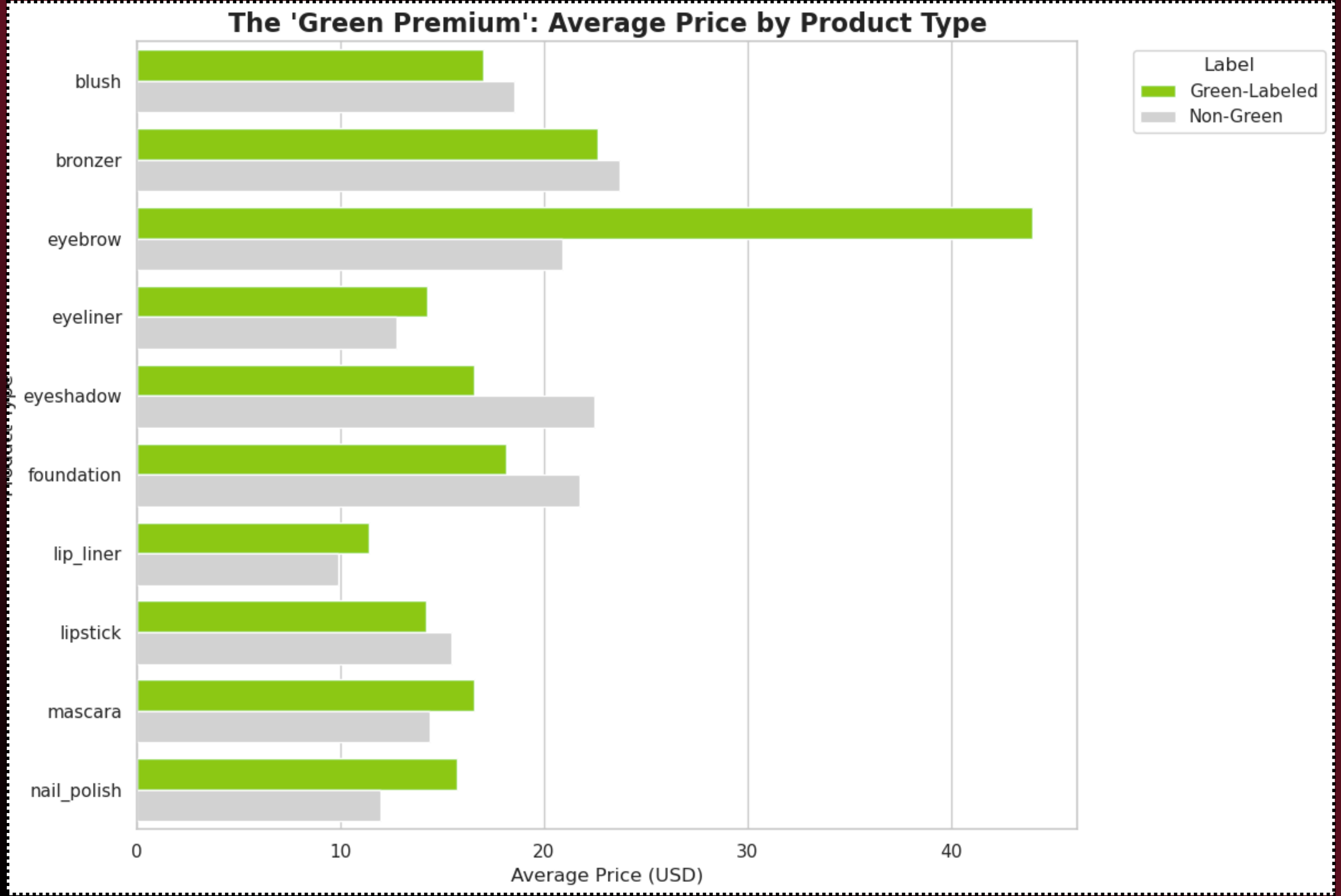
Finding II: The Popularization of Green Beauty

INCONSISTENT PREMIUM

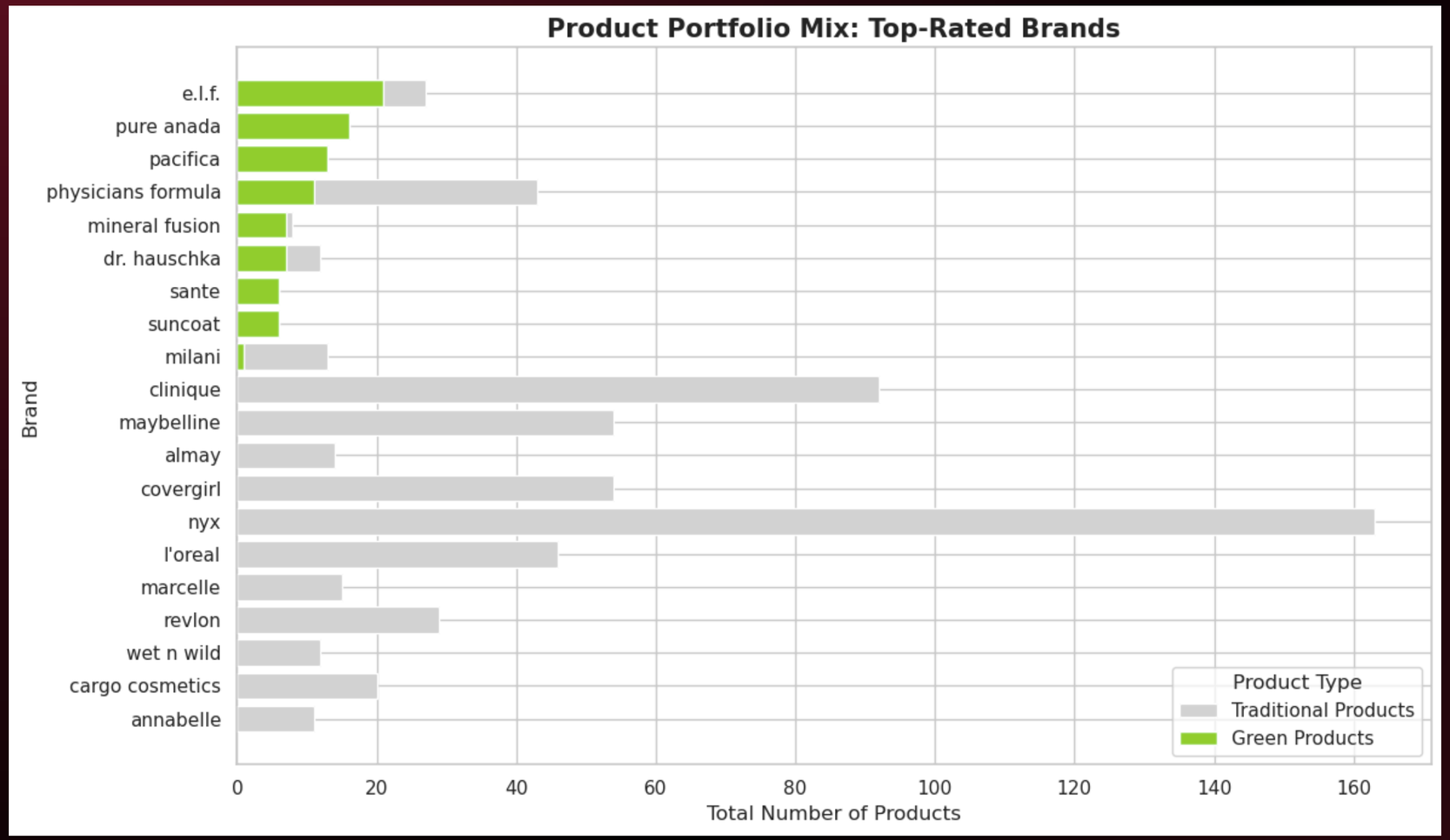
THE PRICE-DROP PHENOMENON

MARKET MATURITY




The 'Green Premium': Average Price by Product Type
Label
Green-Labeled
Non-Green
blush
bronzer
eyebrow
eyeliner
eyeshadow
foundation
lip_liner
lipstick
mascara
nail_polish
0
10
20
30
40
Average Price (USD)

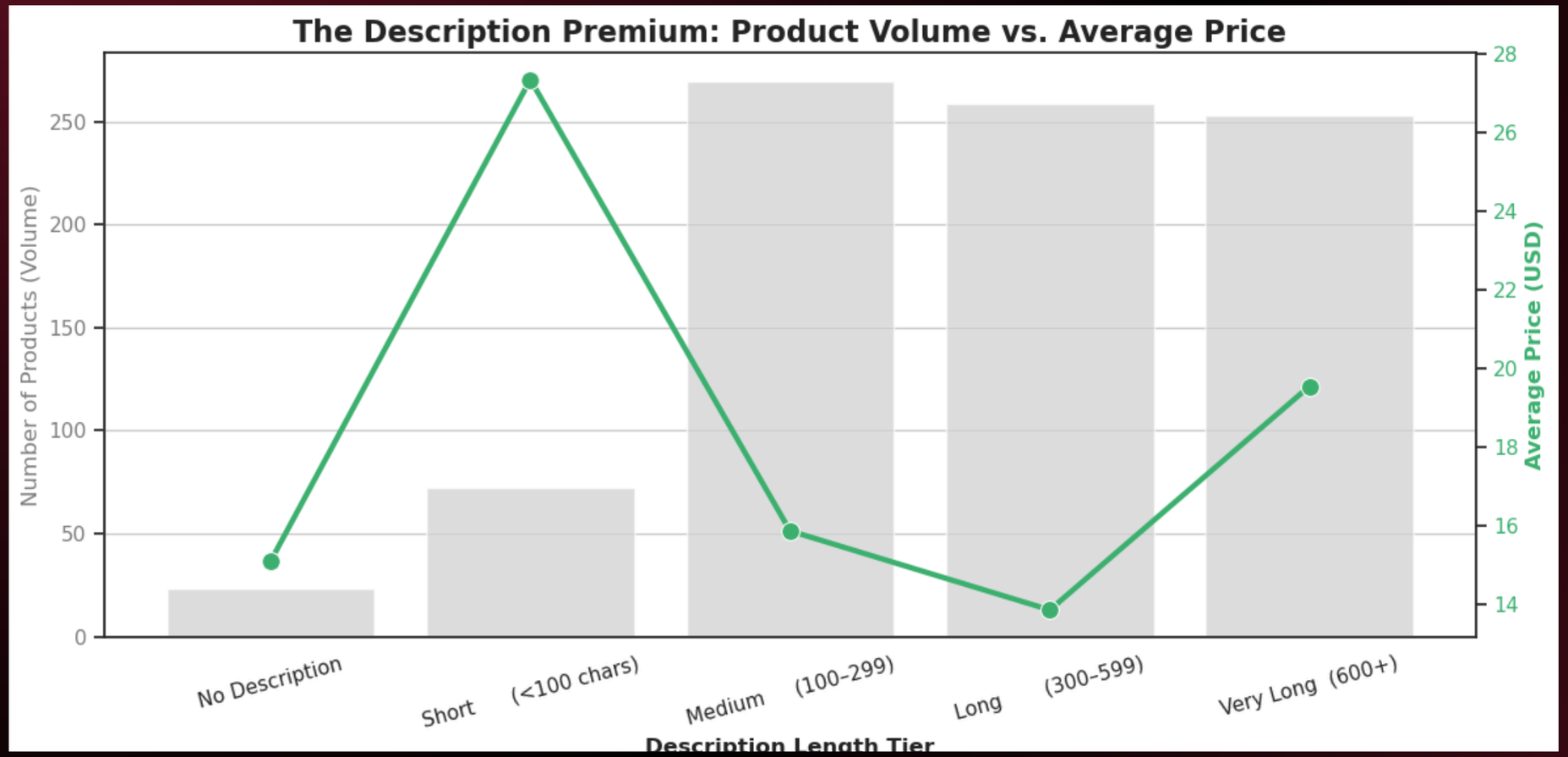
Insight: Green Product Brand Portfolios



100% Green Portfolios are fully viable

Brand Equity and Specialized Positioning

Finding III: The Description Premium



Luxury Relies on Brand Confidence

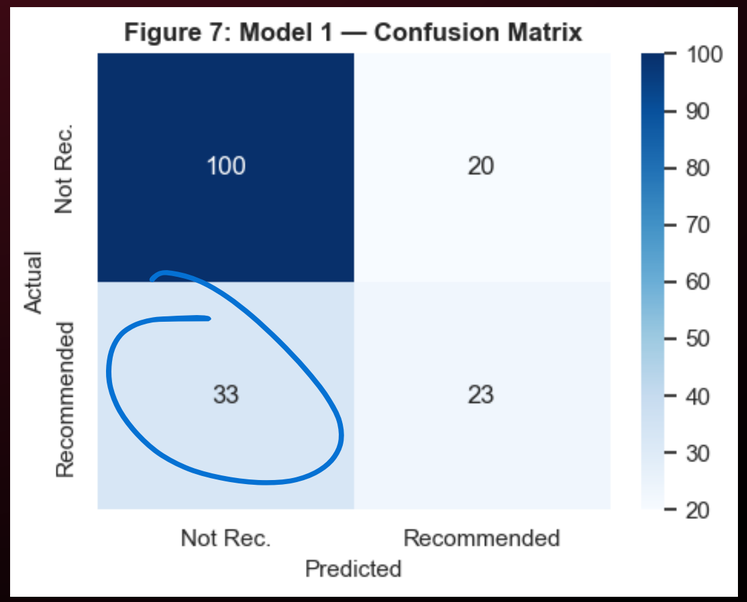
Short Descriptions Paradox

The Mid-Market "Romance"

Mid-Market relies on Copy Complexity

Model I: Recommended Products

	precision	recall
Not Recommended	0.75	0.83
Recommended	0.53	0.41
accuracy		
macro avg	0.64	0.62
weighted avg	0.68	0.70



- PRICE-QUALITY MYTH EVIDENCE:
  - PRECISION (0.68) VS RECALL (0.70). A 2% Difference!
- MODEL IDENTIFIES 'LOSERS' BETTER THAN 'WINNERS'
- CAN'T PREDICT CONSUMER SATISFACTION, AS IT IS NON-LINEAR, HIGH-DIMENSIONAL
- THIS CONFIRMS THAT PRICE & WORD COUNT ARE WEAK SIGNALS OF WHAT CONSUMERS LIKE.

Models II & III: Predicting Price

WHAT ACTUALLY EXPLAINS WHY SOME PRODUCTS ARE MORE EXPENSIVE THAN OTHERS?

- OUR INTIAL R^2 WAS VERY LOW (USING RATING AND WORD COUNT), SO WE DECIDED TO ADD BRAND AVERAGE PRICE AND BRAND PRODUCT COUNT AS PREDICTORS.

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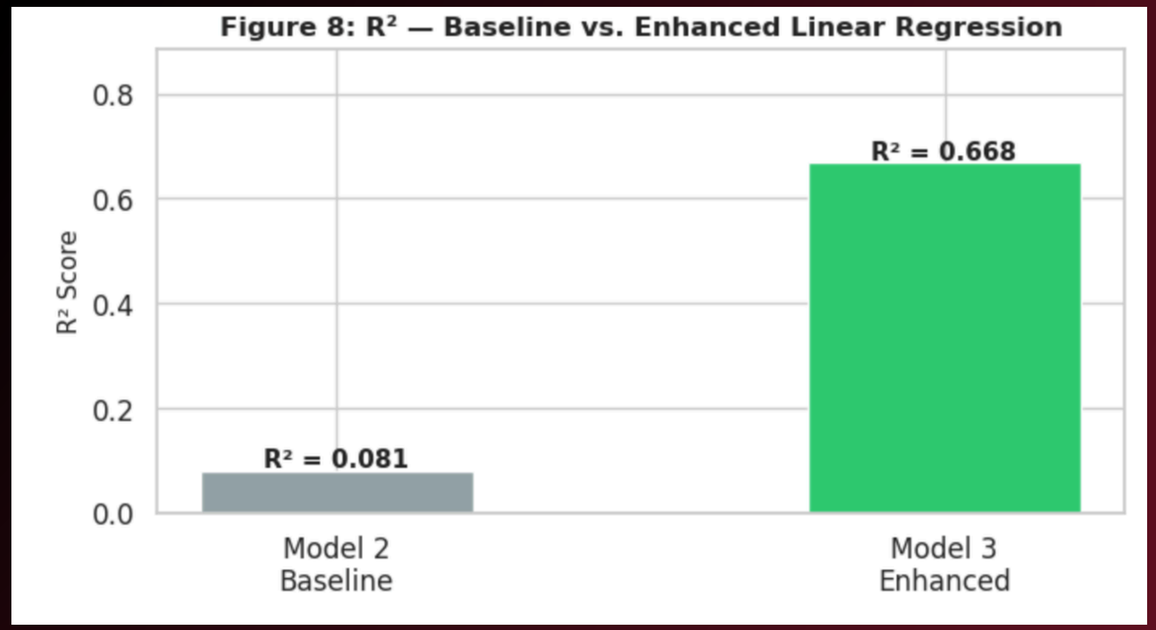
=== Model 3: Enhanced Linear Regression ===
Features used : 21
R²  (test)    : 0.7080
Adjusted R²   : 0.6682
MAE           : $3.56
RMSE          : $6.08

Adj. R² improvement over baseline: +0.5868
```

- WE FOUND THAT BRAND POSITIONING IS THE PRIMARY DRIVER OF PRICE, FAR OUTWEIGHING RATING & DESCRIPTION LENGTH
  - SEVERELY BOOSTED OUR ADJUSTED R^2: 66.8% VARIANCE EXPLAINED

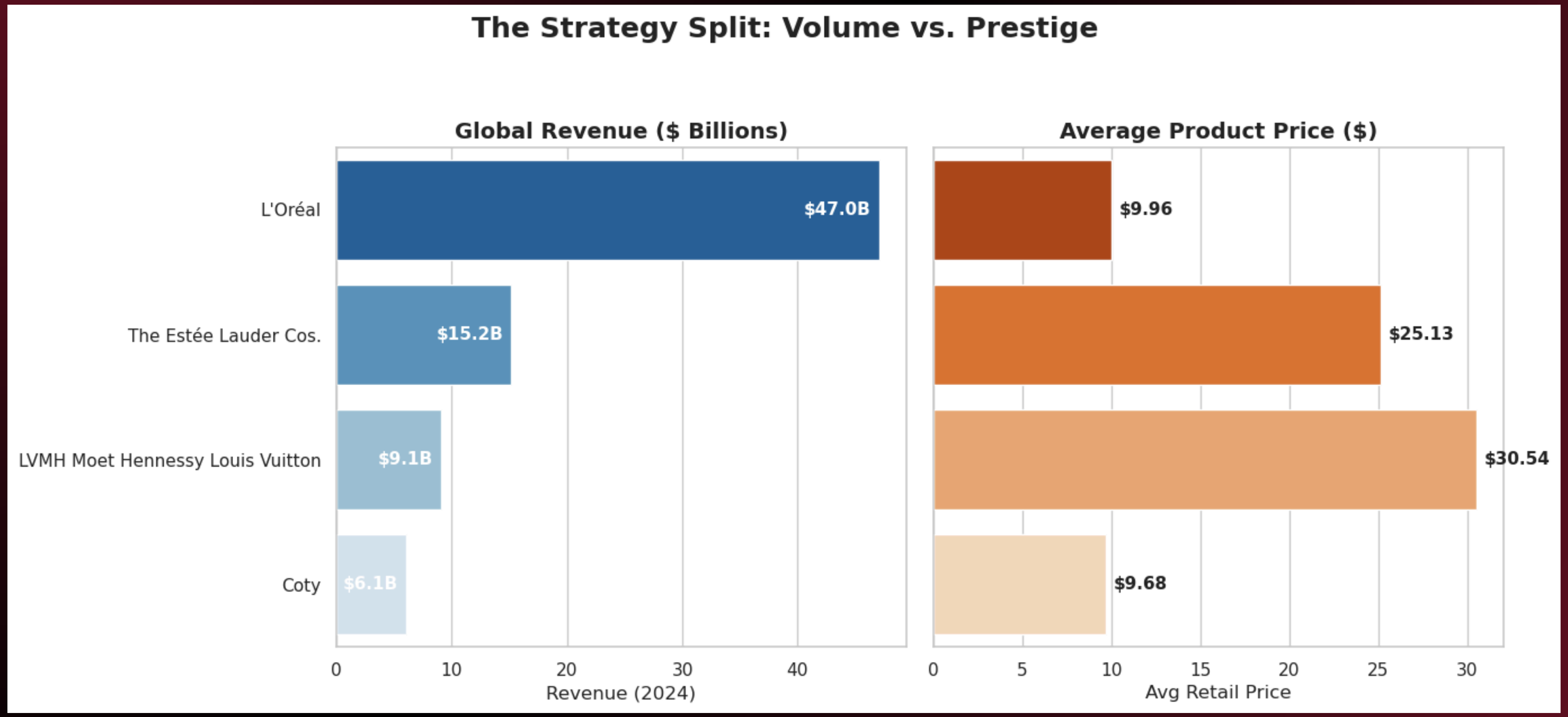


Recap: Marketing Strategy Insights

OUR MODELS CONFIRMED OUR FINDINGS!

- LUXURY BRANDS RELY ON BRAND PRESTIGE, NOT VERBOSE COPY
  - IT IS ALL ABOUT EXISTING BRAND EQUITY
- CLEAN BEAUTY IS A BASELINE REQUIREMENT FOR NEW COMPANIES & MARKET ENTRY
- TOP-RATED BRANDS ARE MID-SIZED, CONSUMER REWARD PRODUCT SPECIALIZATION NOT GENERALISTS

The Success of Budget-Friendly Brands



Next Steps

1. DO "SHORT-DESCRIPTION" HIGH-PRICE ITEMS ALL BELONG TO SPECIFIC LUXURY BRANDS?

2. SENTIMENT ANALYSIS TO LOOK FOR WHAT GREEN PRODUCTS GET MORE "EMOTIONAL/POSITIVE" REVIEWS, NOT JUST RATINGS.

3. REGRESSION MODEL: GIVEN THE COMPLEX NATURE OF RATINGS IN THE COSMETIC INDUSTRY, ADDITIONAL FEATURES (SUCH AS INGREDIENTS) AND A MORE COMPLEX MODEL (SUCH AS RANDOM FOREST) COULD BE USED TO TEST OVERLAPPING MARKERS.

Thank You!